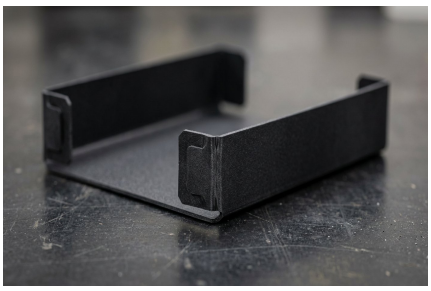
SHIELD SERIES (Shield-FGK / Shield-NMX)

InsulTherm Shield™

Crease-Scored 3D Dielectric Busbar Origami & High-Voltage Isolation System

InsulTherm Shield™ is a precision digital CNC kiss-cut and 50%-depth crease-scored 3D dielectric insulation system engineered to replace expensive injection-molded plastic busbar shrouds and hard-stamped liners. Converted from flame-retardant polypropylene (Formex™ GK equiv.) or high-temperature calendered aramid/PET duplex laminate (Nomex® 410 equiv.), Shield™ ships flat and snap-folds by hand in <5 seconds via zero-memory living hinges around complex 3D copper busbars, collector plates, and 1,500V DC battery rack trays.



Macro Cross-Section / Studio Hardware Inspection

UL 94 V-0	>22–32 kV/mm	1,500V DC Rated	₹0 Tooling NRE	100% India DVA

1. CHEMICAL / MATERIAL COMPOSITION & INTERNAL LAYER ARCHITECTURE

Monolithic extruded flame-retardant polypropylene sheet (Shield-FGK) or high-density calendered aromatic polyamide / PET duplex composite (Shield-NMX). Converted on a multi-axis vacuum-bed digital oscillating-knife table with depth-controlled scoring (50% ± 3% wall depth) that aligns polymer chains along the fold axis, preventing stress-whitening or dielectric loss.

Top Dielectric Face	Corona-treated matte flame-retardant polypropylene / calendered meta-aramid surface
50% CNC Living Hinge	Precision depth-controlled score line enabling 90°–180° 5-second snap folding
Core Isolation Matrix	0.25–0.80 mm non-hygroscopic high-dielectric polymer matrix (>22–32 kV/mm)
Optional PSA Strip	Selective UL 94 V-0 acrylic mounting strip with finger-lift PET liner where specified

2. TYPICAL PHYSICAL, THERMAL & ELECTRICAL PROPERTIES

PROPERTY / PARAMETER	TYPICAL ENGINEERING VALUE	TEST METHOD / STANDARD
Dielectric Breakdown Voltage	**> 22 – 32 kV/mm (11 – 16 kV @ 0.50 mm)**	ASTM D149 / IEC 60243
Flammability Rating	**UL 94 V-0 (Non-dripping, self-extinguishing)**	UL 94 Vertical Burn
Standard Thickness Range	**0.25 mm – 0.80 mm (0.40 mm & 0.43 mm stock)**	ASTM D374 Micrometer
Continuous Operating Temp	**-40°C to +115°C (FGK) / -60°C to +205°C (NMX)**	UL 746B / IEC 60216
Volume Resistivity	**> 1.0 × 10^16 Ω·cm**	ASTM D257
Moisture Absorption (24h)	**< 0.06% (Shield-FGK hydrophobic matrix)**	ASTM D570
Thermal Conductivity	**0.14 – 0.16 W/m·K (Electrical & arc barrier)**	ASTM C518
Comparative Tracking Index	**> 600 V (PLC 0)**	IEC 60112
Living-Hinge Score Depth	**50% ± 3% thickness; >1,000 fold cycles**	InsulTherm QC-CNC-04
System Voltage Architecture	**Up to 1,500V DC BESS, Solar & Rail Busbars**	IEC 60664-1 Clearance

InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India

Engineering Notice: Typical laboratory & baseline metrics (ASTM D5470 / UL 94 / ASTM E595). System-level OEM qualification is required prior to volume release.

3. ORDERING NOMENCLATURE, PREMIX VARIANTS & COMMERCIAL PRICING SCHEDULE

PART / GRADE CODE	MATERIAL SPECIFICATION & CONSTRUCTION	PROTOTYPE / R&D TIER	VOLUME / OEM TIER
`Shield-FGK-043`	Flame-Retardant Polypropylene (Formex™ GK-17 equiv.), 0.43 mm, Matte Black	**₹45 / folded pc (1–50 pcs)**	**₹30 / pc (Volume)**
`Shield-NMX-050`	High-Temp Aramid / PET Duplex (Nomex® 410 equiv.), 0.50 mm, Amber/Natural	**₹45 / folded pc (1–50 pcs)**	**₹30 / pc (Volume)**
`Custom CAD Profile`	24–48h Digital CNC Cut & Score from .DXF / .STEP (Zero NRE Tooling Fee)	**₹0 NRE | MOQ: 1 pc**	**Rotary Toll >10k/mo**

4. TARGET INDUSTRIAL APPLICATIONS & DEPLOYMENT SECTORS

- 1,500V DC solar string inverters, industrial Variable Frequency Drives (VFDs), and UPS collector isolation.
- Railway propulsion traction converters (Vande Bharat / Kavach signalling power enclosures) replacing ₹25,000–₹60,000 plastic molds.
- Grid-scale BESS 20-ft container steel tray dielectric liners and high-amperage laminated copper busbar shrouds.
- EV traction battery pack high-voltage terminal post shields and BDU (Battery Disconnect Unit) arc-flash dividers.

5. DIGITAL CNC CONVERTING, APPLICATION & STORAGE GUIDELINES

1. CAD Submission: Supply flat-pattern .DXF or 3D sheet-metal .STEP files (specify K-factor = 0.40 and bend radius = 0.5× thickness). 2. Assembly: Pre-scored parts ship flat in anti-static sleeves. Fold briskly past 90° along CNC crease lines to engage integrated tab-slot mechanical locks in <5 seconds without adhesives or heat guns. 3. Storage: Store flat at 15°C–35°C, <75% RH away from direct UV exposure. Shelf life: 60 months (uncoated) / 24 months (PSA-backed).

6. QUALITY ASSURANCE, REGULATORY COMPLIANCE & SOVEREIGN DVA TRACEABILITY

Thermal & Electrical Metrology	Batch-wise ASTM D5470-17 thermal impedance calibration curves (10–50 psi) & ASTM D149 dielectric verification.
Flammability & Fire Safety	UL 94 V-0 rated material formulations; supports ARAI AIS-156 Amendment 3 & UL 9540A pack/rack propagation testing.
Outgassing & Environmental	RoHS 3 (EU 2015/863) & REACH compliant; ASTM E595 low-outgassing trace on Graphite-Flex™ & Ultra-NS™ grades.
Domestic Value Addition (DVA)	100% India Digital CNC Converting (Gujarat); compliant with PM E-DRIVE, SECI BESS DVA & <₹200 Cr MoD GTE Ban.

RAPID 24–48 HOUR CAD-TO-PART PROTOTYPING & OEM PROCUREMENT DESK
Submit .DXF / .STEP drawings, request <₹15,000 Petty-Cash R&D Sampler Kits, or schedule an engineering review:
Primary Official Desk: manav.sb@inslthrm.in | **Active Operations & Order Routing:** manavsb9719@gmail.com
Engineering Helpline: +91 78148 XXXXX | **Manufacturing Wing:** InsulTherm Innovations, Advanced Materials & Digital Converting Wing, Block C-14, Sanand GIDC Industrial Estate (Near iCreate Campus), Ahmedabad, Gujarat – 382110, India

Engineering Notice: Typical laboratory & baseline metrics (ASTM D5470 / UL 94 / ASTM E595). System-level OEM qualification is required prior to volume release.